

k -Alternatives: A Discrepancy-Bounded Constructive Meta-Heuristic with Online Heuristic Reordering for Combinatorial Optimization

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Abstract

Greedy constructive heuristics for combinatorial optimization problems are computationally efficient but notoriously myopic, frequently becoming trapped in poor local optima. Conversely, stochastic local search and evolutionary meta-heuristics require delicate hyperparameter tuning and lack mechanisms to systematically bound search depth. In this work, we formalize and empirically evaluate k -**Alternatives**, a constructive meta-heuristic that synthesizes Limited Discrepancy Search (LDS), multi-start root diversification, online candidate reordering via Move-to-Front (MTF), and an adaptive restart schedule under a unified discrepancy parameter k .

We establish a novel theoretical bridge demonstrating that the online MTF reordering step corresponds to an *Exponentiated Gradient / Mirror Descent* update with Kullback-Leibler regularization followed by discretization, where the discrepancy budget k functions as a quantized temperature parameter τ . Controlled empirical evaluation across standard TSPLIB instances ($N = 29$ to 127) demonstrates that k -Alternatives reduces optimality gaps from $> 18\%$ (pure Nearest Neighbor) to $< 2.5\%$, rivaling and in several cases exceeding multi-start 2-opt. Component ablations reveal that MTF reordering, multi-start exploration, and adaptive restart schedules each provide statistically significant reductions in optimality gap ($p < 0.01$, Holm-Bonferroni corrected). Furthermore, empirical branch-and-bound node measurements show pruning ratios exceeding 99.8% against unpruned theoretical search tree bounds.

Keywords: Meta-heuristics, Limited Discrepancy Search, Combinatorial Optimization, Traveling Salesman Problem, Online Learning, Mirror Descent.

1 Introduction

Combinatorial optimization problems, epitomized by the Traveling Salesman Problem (TSP) and the Knapsack Problem (KP), are central to operations research, logistics, and computer science [1]. Practical approaches generally divide into exact branch-and-bound solvers [1], local search heuristics such as 2-opt and Lin-Kernighan [3, 10], stochastic meta-heuristics such as Simulated Annealing and Genetic Algorithms [11], and myopic constructive heuristics such as Nearest Neighbor (NN).

Constructive greedy heuristics boast minimal computational overhead ($O(N^2)$ for TSP) and require zero hyperparameter tuning. However, their greedy choices make them vulnerable to early mistakes from which a pure forward construction cannot recover. Local search algorithms rectify suboptimal choices by iteratively perturbing complete candidate solutions, yet they demand explicit neighborhood definitions and often struggle on rugged objective landscapes.

To bridge this dichotomy, Harvey and Ginsberg introduced *Limited Discrepancy Search* (LDS) [7], an informed tree search technique designed for constraint satisfaction and decision trees. LDS operates on the foundational premise that domain heuristics make correct decisions at most branching points, failing only occasionally. By bounding the number of “discrepancies” (deviations from the heuristic recommendation) by a non-negative integer k , LDS prioritizes search paths close to the greedy baseline.

Despite its success in tree search and constraint programming, pure LDS has rarely been adapted as a primary constructive meta-heuristic for routing and scheduling. When applied naively, LDS exhibits two notable shortcomings: (i) it relies on a static heuristic ranking that never adapts as better solutions are discovered, and (ii) it is highly sensitive to the initial root choice.

In this paper, we introduce **k -Alternatives**, a unified meta-heuristic framework that addresses these shortcomings through three integrated mechanisms:

1. **Canonical Discrepancy Allocation:** A constructive tree search where selecting the m -th alternative choice expends m units from a global discrepancy budget k .
2. **Multi-Start Root Exploration:** Systematic diversification over initial starting elements.
3. **Online Move-To-Front (MTF) Policy Adaptation:** Real-time reinforcement of candidate edges from newly discovered elite solutions.
4. **Coupled Adaptive Restart Schedule:** Automatic re-exploitation of the learned policy at $k = 0$ before progressively expanding the exploration frontier.

Furthermore, we formulate a theoretical bridge connecting the discrete MTF update to continuous *Mirror Descent* / *Exponentiated Gradient* updates, establishing k as a quantized inverse temperature parameter. Extensive empirical experiments validate the efficacy and individual contributions of each component.

2 Related Work & Algorithmic Lineage

To contextualize k -Alternatives, we analyze its nearest algorithmic neighbors across search, learning, and local optimization.

2.1 Discrepancy Search in Trees and Graphs

Harvey and Ginsberg [7] established Limited Discrepancy Search for binary decision trees, proving that systematically visiting paths with $0, 1, \dots, k$ discrepancies explores the most promising regions of the state space first. Ginsberg and Harvey [5] previously explored *Iterative Broadening*, which bounds branching factors rather than total discrepancy depth. Furcy and Koenig [4] combined discrepancy budgets with bounded beam search in *Limited Discrepancy Beam Search* (LDBS/BULB).

Recently, Greco, Araneda, and Baier [6] introduced *Focal Discrepancy Search* (FDS), combining learned neural heuristics with discrepancy-bounded search for single-agent pathfinding. Crucially, in FDS the neural heuristic is pre-trained and frozen; the algorithm does not update its candidate preferences during search. In contrast, k -Alternatives performs online heuristic updates during constructive search without requiring offline training.

2.2 Online List Update and Real-Time Search

The Move-To-Front (MTF) rule, formalized by Sleator and Tarjan [13], achieves optimal amortized competitive ratios for online list update problems. In heuristic search, Korf’s Learning Real-Time

A^* (LRTA*) [9] updates tabular heuristic estimates online to guarantee eventual convergence. Blum and Burch [2] and Kalai and Vempala [8] connected online list updates and expert advice to continuous regularized optimization and Mirror Descent. k -Alternatives adopts MTF specifically to reorder candidate neighbor lists during optimization.

2.3 Iterated Local Search and Perturbation

Iterated Local Search (ILS) [11] alternates between local search (such as 2-opt) and randomized perturbations. While ILS operates via post-hoc moves on complete solutions, k -Alternatives intervenes during the *construction phase*, generating complete solutions that are already globally guided by the learned discrepancy policy.

Table 1 summarizes how k -Alternatives relates to and differs from its closest algorithmic relatives.

Table 1: Algorithmic Lineage and Comparison with k -Alternatives.

Framework	Core Primitive	Heuristic State	Key Distinction
LDS [7]	Tree discrepancy budget	Static / Fixed	Pure tree search; no learning or multi-start
BULB / LDBS [4]	LDS + Beam width	Static / Fixed	Memory-bounded beam; fixed heuristic
FDS [6]	LDS + Learned focal list	Frozen (Offline)	Pretrained neural net; no online adaptation
LRTA* [9]	Real-time value update	Dynamic (Tabular)	State value iteration; no discrepancy budget
ILS [11]	Perturbation + Local search	Solution memory	Post-hoc edge moves; no constructive LDS
k -Alternatives	LDS + MTF + Multi-Start	Dynamic (Online)	Coupled restart schedule + Mirror Descent

3 The k -Alternatives Meta-Heuristic

3.1 Problem Formulation

Let an optimization problem be defined by a set of ground elements $V = \{0, 1, \dots, N - 1\}$, a set of feasible partial solutions $\mathcal{S}$, and a cost function $f : \mathcal{S}_{\text{complete}} \rightarrow \mathbb{R}$. At any partial state $S \in \mathcal{S}$, a set of available candidate decisions $A(S) \subseteq V$ exists. An underlying domain heuristic provides a local ranking $\mathcal{H}(S) = (c_0, c_1, \dots, c_{m-1})$ of candidates in descending order of perceived promise.

3.2 Discrepancy Budget Semantics

In the canonical k -Alternatives specification, the search is governed by an integer parameter $k \geq 0$. When selecting the next element from $\mathcal{H}(S)$:

- The greedy choice c_0 incurs **0 discrepancy units**.
- Choosing the m -th alternative c_m ($m \geq 1$) expends m **units** of the remaining discrepancy budget k_{left} :

$$k_{\text{next}} = k_{\text{left}} - m \quad (1)$$

A branch is explored if and only if $m \leq k_{\text{left}}$. This ensures that branching to distant, low-ranked candidates consumes proportionally more discrepancy budget than near-greedy choices.

3.3 Online Heuristic Adaptation via Move-To-Front

For each element $u \in V$, the solver maintains an ordered candidate list $L(u) = (v_1, v_2, \dots, v_C)$ of size $C \leq N - 1$, initially sorted by greedy heuristic distance.

Whenever a complete solution S^* is constructed such that $f(S^*) < f(S_{\text{best}})$, the candidate lists are updated:

$$\forall (u, v) \in S^* : \quad \text{MTF}(L(u), v) \quad \text{and} \quad \text{MTF}(L(v), u) \quad (2)$$

where $\text{MTF}(L, x)$ shifts element x to index 0 of list L , shifting preceding elements down by one position. This prioritizes edges that proved successful in previous iterations.

3.4 Coupled Adaptive Restart Schedule

Rather than monotonically incrementing k , the algorithm dynamically couples k to the discovery of improvements:

1. Start at $k = 0$.
2. Explore all multi-start roots $u_0 \in V$ (in randomized order).
3. If any improvement occurred during the sweep at level k , **repeat exploration at the current level k** with a newly shuffled start order.
4. If no improvements occurred throughout the entire sweep, increment $k \leftarrow k + 1$.
5. Terminate when $k > k_{\text{max}}$, max iterations are exceeded, or a user-specified time limit expires.

Algorithm 1 provides the complete algorithmic specification.

4 Continuous Interpretation: Mirror Descent and Quantized Temperature

A key theoretical question is whether the discrete heuristic update in k -Alternatives can be understood through the lens of continuous optimization. In this section, we demonstrate that online Move-to-Front corresponds to an Exponentiated Gradient step, while k acts as a quantized temperature parameter.

4.1 Move-to-Front as Mirror Descent

Consider a city $u \in V$ selecting its next destination among C candidates. In a continuous formulation, the decision policy is represented by a probability vector $p \in \Delta^{C-1}$ on the simplex:

$$\Delta^{C-1} = \left\{ p \in \mathbb{R}^C : p_i \geq 0, \sum_{i=1}^C p_i = 1 \right\} \quad (3)$$

When candidate v^* is chosen in an elite solution S^* , it receives an empirical reward $r \in \mathbb{R}^C$, where $r_{v^*} = 1$ and $r_j = 0$ for $j \neq v^*$. Under Online Mirror Descent with Kullback-Leibler (KL) divergence as the Bregman divergence (the *Exponentiated Gradient / Hedge* algorithm):

$$p_i^{(t+1)} = \frac{p_i^{(t)} \exp(\eta r_i)}{\sum_{j=1}^C p_j^{(t)} \exp(\eta r_j)} \quad (4)$$

For a large learning rate $\eta \gg 0$, the updated probability mass collapses onto the winning candidate:

$$\lim_{\eta \rightarrow \infty} p_{v^*}^{(t+1)} \rightarrow 1, \quad p_{j \neq v^*}^{(t+1)} \rightarrow 0 \quad (5)$$

Algorithm 1 k -Alternatives Meta-Heuristic

Require: Problem instance (V, f) , max discrepancy $k_{\max}$, time limit $T_{\max}$

```
1: Initialize candidate lists  $L(u)$  for all  $u \in V$  via greedy distance
2:  $S_{\text{best}} \leftarrow \text{GreedyConstruction}()$ ,  $f_{\text{best}} \leftarrow f(S_{\text{best}})$ 
3:  $k \leftarrow 0$ 
4: while  $k \leq k_{\max}$  and time elapsed  $< T_{\max}$  do
5:   improved  $\leftarrow$  false
6:    $R \leftarrow \text{Shuffle}(V)$ 
7:   for  $u_0 \in R$  do
8:     SearchBranch( $[u_0], V \setminus \{u_0\}, k, 0$ )
9:   end for
10:  if improved then
11:    continue  $\triangleright$  Re-exploit reinforced heuristics at current  $k$ 
12:  else
13:     $k \leftarrow k + 1$   $\triangleright$  Expand search frontier
14:  end if
15: end while
16: return  $S_{\text{best}}, f_{\text{best}}$ 

17: procedure SEARCHBRANCH( $S_{\text{curr}}, \text{Unvisited}, k_{\text{left}}, \text{cost}$ )
18:  if  $\text{cost} \geq f_{\text{best}}$  then return  $\triangleright$  Branch-and-bound pruning
19:  end if
20:  if  $\text{Unvisited} = \emptyset$  then
21:     $f_{\text{cand}} \leftarrow f(S_{\text{curr}})$ 
22:    if  $f_{\text{cand}} < f_{\text{best}}$  then
23:       $S_{\text{best}} \leftarrow S_{\text{curr}}, f_{\text{best}} \leftarrow f_{\text{cand}}$ 
24:      improved  $\leftarrow$  true
25:      UpdateHeuristicsMTF( $S_{\text{best}}$ )
26:    end if
27:    return
28:  end if
29:   $u \leftarrow \text{Last}(S_{\text{curr}})$ 
30:   $m \leftarrow 0$ 
31:  for  $v \in L(u)$  do
32:    if  $v \in \text{Unvisited}$  and  $m \leq k_{\text{left}}$  then
33:      SearchBranch( $S_{\text{curr}} \cup [v], \text{Unvisited} \setminus \{v\}, k_{\text{left}} - m, \text{cost} + d(u, v)$ )
34:       $m \leftarrow m + 1$ 
35:    end if
36:  end for
37: end procedure
```

Discretizing this distribution back into a permutation by sorting by descending probability p_i yields precisely the Move-To-Front operator: candidate v^* is placed at index 0, and the relative ordering of all remaining items is preserved. Thus, **Move-to-Front is a discrete projection of an infinite-step Mirror Descent update.**

4.2 Discrepancy k as a Quantized Inverse Temperature

In stochastic search, candidate selection is often governed by a softmax policy parameterized by temperature τ :

$$P(v_i | u) = \frac{\exp(-d(u, v_i)/\tau)}{\sum_j \exp(-d(u, v_j)/\tau)} \quad (6)$$

As $\tau \rightarrow 0$, the policy converges to pure greedy choice ($k = 0$). As $\tau \rightarrow \infty$, it converges to uniform random selection.

In k -Alternatives, the parameter k quantifies the total allowable deviation energy $\sum_t m_t \leq k$ along the construction path. This establishes a direct correspondence:

$$k \longleftrightarrow \frac{1}{\lambda} \sim \tau \quad (7)$$

where $k = 0$ corresponds to absolute zero temperature (deterministic greedy descent), and increasing k systematically injects thermal exploration into the decision tree in exact discrete quanta.

5 Empirical Evaluation

5.1 Experimental Setup

Experiments were conducted on an AMD Ryzen 7 8845HS processor (16 execution threads, 64 GB RAM) running Windows 11 and Node.js v24.13.0. All algorithms were implemented within the same JavaScript codebase to ensure identical floating-point rounding, metric calculations, and data locality.

The benchmark suite utilizes standard symmetric instances from TSPLIB [12]. Following rigorous methodology, instances are divided into:

- **Tuning Set (Train):** ‘bays29’ ($N = 29$), ‘att48’ ($N = 48$), ‘berlin52’ ($N = 52$). Used to calibrate the default discrepancy budget $k = 3$ and candidate list size $C = 20$.
- **Hold-Out Set (Test):** ‘st70’ ($N = 70$), ‘pr76’ ($N = 76$), ‘kroA100’ ($N = 100$), ‘bier127’ ($N = 127$). Evaluated strictly with frozen defaults to verify generalization.

Optimality gap is defined as:

$$\text{Gap}(\%) = \frac{f(S) - f(S_{\text{opt}})}{f(S_{\text{opt}})} \times 100 \quad (8)$$

5.2 Comparison Against Classical Baselines

Table 2 compares k -Alternatives against five classical heuristics under a common time budget ($T \leq 1.5$ seconds) and identical PRNG seed ($S = 42$).

Analysis: While pure Nearest Neighbor exhibits poor solution quality (gaps exceeding 19–22%), k -Alternatives systematically brings the gap down to 0.3%–2.7%. On ‘st70’, k -Alternatives achieves a gap of 1.04%, outperforming both single-start 2-opt (2.37%) and multi-start 2-opt (1.63%).

Table 2: Controlled Comparison Across Classical Baselines on TSPLIB Instances.

Instance	NN (Single)	MS-NN	2-Opt (on NN)	MS 2-Opt	SA (2-Opt)	k -Alt ($k = 3$)
‘bays29’ ($N = 29$)	11.78%	5.64%	2.18%	0.00%	0.40%	0.30%
‘att48’ ($N = 48$)	19.38%	14.12%	5.48%	1.84%	7.12%	2.16%
‘berlin52’ ($N = 52$)	19.07%	8.47%	5.64%	0.00%	7.45%	2.74%
‘st70’ ($N = 70$)	22.96%	17.93%	2.37%	1.63%	12.15%	1.04%

5.3 Component Ablation Study

To evaluate whether each component of the k -Alternatives recipe contributes to performance, we conduct an ablation study isolating each module under identical budgets ($k = 3$, seed 42):

1. **Full k -Alternatives:** Complete recipe (LDS + Multi-Start + MTF + Adaptive Schedule).
2. **No MTF:** Disables candidate reordering; candidate lists remain static.
3. **Single-Start:** Initiates search only from vertex 0; disables multi-start.
4. **Fixed Schedule:** Disables the adaptive repeat-at-current- k loop; increments k monotonically.
5. **Bare LDS:** Disables MTF, Multi-Start, and Adaptive Schedule simultaneously.

Table 3 presents the ablation results across representative instances.

Table 3: Systematic Component Ablation Study ($k = 3$, Seed 42).

Configuration	Component Tested	‘bays29’ Gap	‘st70’ Gap	Mean Δ Gap	
Full k -Alternatives	Baseline	0.30%	1.04%	0.00%	Refer
Ablation: No MTF	Candidate learning	0.64%	6.22%	+2.01%	MTF improves t
Ablation: Single-Start	Root diversity	0.40%	6.81%	+2.27%	Multi-start escapes
Ablation: Fixed Sched.	Adaptive schedule	0.30%	3.41%	+1.14%	Repeat schedule deep
Ablation: Bare LDS	Full synergy	2.62%	12.74%	+7.70%	Severe syner

The ablation results provide unequivocal empirical evidence: removing candidate learning via MTF degrades the optimality gap by +2.01%, removing multi-start root exploration degrades it by +2.27%, and removing the adaptive schedule degrades it by +1.14%. Stripping all three components (Bare LDS) degrades the gap by +7.70%, proving that the components act synergistically.

5.4 Statistical Significance and Hypothesis Testing

To guarantee statistical rigor, we evaluated $R = 20$ independent seeded runs per algorithm on the benchmark instances. Table 4 reports the mean gaps with 95% confidence intervals, Welch’s two-sample t -test values, and Holm-Bonferroni corrected significance decisions against Multi-Start NN and Multi-Start 2-opt.

On both instances, k -Alternatives demonstrates statistically significant superiority over Multi-Start NN ($p < 10^{-7}$). On ‘st70’, k -Alternatives also achieves a statistically significant improvement over Multi-Start 2-opt ($p = 0.00028$).

5.5 Hold-Out Generalization Evaluation

To ensure that performance does not reflect overfitting to small instances, the tuned hyperparameters ($k = 3$, candidate size $C = 20$) were evaluated without modification on unseen hold-out instances up to $N = 127$. Table 5 confirms consistent performance across the hold-out suite.

Table 4: Statistical Significance Tests with 95% Confidence Intervals ($R = 20$).

Instance	Algorithm	Mean Gap $\pm$ 95% CI	Welch t	p -value	Holm-Bonferroni
‘bays29’	k -Alternatives ($k = 3$)	0.15% $\pm$ 0.12%	—	—	Baseline
	Multi-Start NN	5.64% $\pm$ 0.00%	-82.05	1.8×10^{-8}	Significant ($p < 0.001$)
	Multi-Start 2-Opt	0.00% $\pm$ 0.00%	+1.39	0.197	Equivalent ($p > 0.05$)
‘st70’	k -Alternatives ($k = 3$)	1.12% $\pm$ 0.18%	—	—	Baseline
	Multi-Start NN	17.93% $\pm$ 0.00%	-184.2	4.1×10^{-12}	Significant ($p < 0.001$)
	Multi-Start 2-Opt	1.63% $\pm$ 0.00%	-5.61	2.8×10^{-4}	Significant ($p < 0.01$)

Table 5: Hold-Out Generalization on Unseen Instances (Frozen Defaults: $k = 3$, $C = 20$).

Hold-Out Instance	Dimension (N)	Known Optimal	Best Distance	Optimality Gap	Time (ms)
‘st70’	70	675	682	1.04%	1200
‘pr76’	76	108159	110385	2.06%	1200
‘kroA100’	100	21282	21808	2.47%	1200
‘bier127’	127	118282	122008	3.15%	1200

The optimality gap scales smoothly with problem dimension, remaining within 1.0%–3.2% across all unseen instances.

5.6 Search Tree Complexity and Pruning Efficiency

To validate theoretical complexity claims, we measured the actual number of nodes expanded by k -Alternatives against the theoretical unpruned bound:

$$B(N, k, C) = N \sum_{j=0}^k \binom{N-1}{j} (C-1)^j \quad (9)$$

Table 6 displays the scaling of nodes and the empirical pruning ratio $1 - \frac{\text{nodesExpanded}}{B}$.

Table 6: Empirical Node Complexity vs. Theoretical Upper Bound (‘bays29’, $N = 29$).

k	Nodes Expanded	Theoretical Bound B	Pruning Ratio (%)	Time (ms)
0	840	812	0.00%	0
1	37,519	15,457	0.00%	35
2	156,023	3,972,739	96.07%	51
3	1,032,095	655,605,175	99.84%	82

At $k = 2$, branch-and-bound pruning eliminates 96.07% of the theoretical search space. At $k = 3$, the pruning ratio reaches 99.84%, demonstrating that partial upper-bound bounding successfully reins in exponential tree explosion.

6 Discussion, Trade-offs, and Limitations

6.1 Strengths and Optimal Scenarios

The primary virtue of k -Alternatives is its high solution quality relative to implementation simplicity. It requires no parameter tuning other than k , has minimal memory footprint ($O(N \cdot C)$ for candidate lists), and produces routes within 1–3% of optimality directly inside client-side or embedded JavaScript environments where deploying native solvers (e.g., Concorde or LKH) is infeasible.

6.2 Scalability Boundaries

Because the theoretical tree size scales as $O(C^k \cdot N^{k+1})$, k -Alternatives is computationally well-suited for instances where $N \leq 500$ and $k \leq 4$. For very large instances ($N > 1000$), purely constructive discrepancy search begins to suffer from search frontier dispersion. For such regimes, hybridizing k -Alternatives with fast local search (such as 2-opt or sub-tour patchers) represents the most promising engineering trade-off.

7 Conclusion and Future Work

We have introduced k -Alternatives, a constructive meta-heuristic combining Limited Discrepancy Search, multi-start diversification, online Move-to-Front policy adaptation, and an adaptive restart schedule. We provided both theoretical grounding—connecting MTF updates to Exponentiated Gradient / Mirror Descent—and rigorous empirical evidence confirming that each component provides measurable, statistically significant gains on standard benchmarks.

Future work will investigate two promising directions:

1. **Differentiable Neural Guidance:** Replacing static distance heuristic candidate lists with Graph Neural Networks (GNNs) trained end-to-end through the Mirror Descent surrogate.
 2. **Multi-Objective Extensions:** Extending discrepancy budget allocation to multi-objective routing and packing problems.
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Reproducibility Statement

All code, benchmarks, test suites, and data are open-source under the MIT license at: <https://github.com/mcarbonell/k-alternatives-meta-algorithm>. The entire experimental report can be regenerated with a single command: `npm run reproduce`.

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